



The Scots College

Year 12 Mathematics Extension 2

Task 3

10 June 2010

Name: _____

General Instructions

- Working time - 45 minutes
- Write using blue or black pen
- Board approved calculators may be Used (Non Graphic)
- All necessary working should be shown in every question
- Standard Integrals Table attached

TOTAL MARKS: 32

WEIGHTING: 20 %

Question 1 (Marks 17)

- a) Evaluate [2]

$$\int_3^4 \frac{dx}{\sqrt{x^2 - 9}}$$

- b) By completing the square, find [2]

$$\int \frac{1}{x^2 - 10x + 34} dx$$

- c) i) Find the real numbers a , b and c such that [2]

$$\frac{3 - x}{(1 + 2x^2)(1 + 6x)} \equiv \frac{ax + b}{1 + 2x^2} + \frac{c}{1 + 6x}$$

- ii) Hence find

$$\int_0^2 \frac{3 - x}{(1 + 2x^2)(1 + 6x)} dx \quad [3]$$

- d) Find [3]

$$\int (x \log x)^2 dx$$

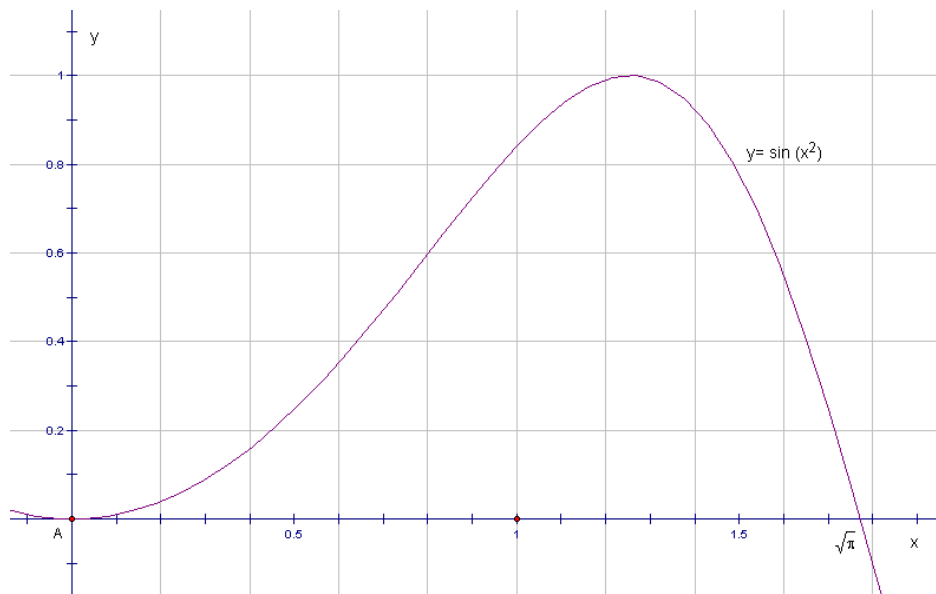
- e) i) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, show that $I_n = \frac{n-1}{n} I_{n-2}$ where $n \in J$ and $n \geq 2$. [5]

- ii) Hence evaluate I_3 .

Question 2 (Marks 17)

- a) A hollow glass container is to be formed by rotating the curve $y = \cos x$ about the line $y = -1$ between $x = 0$ and $x = \pi$. If one unit in the Cartesian plane represents 5 cm, find the volume of liquid it can hold to the nearest millilitre. [5]

- b) Copy the figure below into your worksheet and find the volume of the solid generated by rotating the shaded region about the y-axis. Use the cylindrical shell method. [5]



- c) The base of a particular solid is $x^2 + y^2 = 9$. Find the volume of the solid if every cross-section perpendicular to the x-axis is a semi-ellipse with minor axis in the base of the solid and semi-major axis equal to its minor axis. [5]

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Solution EXT 2 Task 3 June 2010

$$\begin{aligned}
 \text{Q.1. (a)} \quad & \int_3^4 \frac{1}{\sqrt{x^2-9}} dx \\
 &= \int_3^4 \frac{1}{\sqrt{x^2-3^2}} dx \\
 &= \ln(x + \sqrt{x^2-9}) \Big|_3^4 \quad \checkmark \\
 &= \ln(4 + \sqrt{16-9}) - \ln(3 + \sqrt{0}) \\
 &= \ln\left(\frac{4+\sqrt{7}}{3}\right) \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad & \int \frac{1}{x^2-10x+34} dx \\
 &= \int \frac{1}{x^2-10x+25+9} dx \\
 &= \int \frac{1}{(x-5)^2+9} dx \quad \checkmark \\
 &= \frac{1}{3} \tan^{-1}\left(\frac{x-5}{3}\right) + C \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) (i)} \quad & \frac{3-x}{(1+2x^2)(1+6x)} \equiv \frac{ax+b}{1+2x^2} + \frac{c}{1+6x} \\
 & 3-x \equiv (ax+b)(1+6x) + c(1+2x^2)
 \end{aligned}$$

$$\begin{aligned}
 x = -\frac{1}{6} \Rightarrow & c\left(1 + \frac{1}{18}\right) = 3 + \frac{1}{6} \\
 \frac{19}{18}c &= \frac{19}{6} \\
 c &= 3 \quad \#
 \end{aligned}$$

$$\begin{aligned}
 \text{coeff. of } x^2 \quad & 6a + 2c = 0 \\
 & 3a = -c \\
 & 3a = -3 \\
 & a = -1 \quad \#
 \end{aligned}$$

$$\begin{aligned}
 x=0 \Rightarrow & b+c=3 \\
 & b+3=3 \\
 & b=0 \quad \#
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad & \int_0^2 \frac{3-x}{(1+2x^2)(1+6x)} dx \\
 &= \int_0^2 \frac{-x}{1+2x^2} dx + \int_0^2 \frac{3}{1+6x} dx \\
 &= \left[-\frac{1}{4} \ln(1+2x^2)\right]_0^2 + \frac{1}{2} \left[\ln(1+6x)\right]_0^2 \\
 &= -\frac{1}{4} \ln 9 + \frac{1}{4} \ln 1 + \frac{1}{2} \ln 13 - \frac{1}{2} \ln 1 \\
 &= \frac{1}{2} \ln 13 - \frac{1}{2} \ln 9^{\frac{1}{2}} \\
 &= \frac{1}{2} \ln\left(\frac{13}{3}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad & \int (x \log x)^2 dx \\
 &= \int x^2 (\log^2 x) dx \\
 &= (\log x)^2 \frac{x^3}{3} - \int 2(\log x) \cdot \frac{1}{x} \cdot \frac{x^3}{3} dx \quad \checkmark \\
 &= \frac{x^3}{3} (\log x)^2 - \frac{2}{3} \int x^2 \log x dx \\
 &= \frac{x^3}{3} (\log x)^2 - \frac{2}{3} \left[(\log x) \frac{x^3}{3} - \int \frac{1}{x} \cdot \frac{x^3}{3} dx \right] \checkmark \\
 &= \frac{x^3}{3} (\log x)^2 - \frac{2}{9} x^3 \log x + \frac{2}{3} \int x^2 dx \\
 &= \frac{x^3}{3} (\log x)^2 - \frac{2}{9} x^3 \log x + \frac{2}{9} x^3 + C \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (f) \quad I_n &= \int_0^{\pi/2} \cos^n x \, dx \\
 &= \int_0^{\pi/2} \cos^{n-1} x \cdot \cos x \, dx
 \end{aligned}$$

$$\begin{aligned}
 u &= \cos^{n-1} x \quad \checkmark \quad dv = \cos x \, dx \quad \checkmark \\
 \frac{du}{dx} &= -(n-1) \cos^{n-2} x \sin x \quad \checkmark \quad v = \sin x
 \end{aligned}$$

$$= \left[\cos^{n-1} x \cdot \sin x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin x \cdot \sin x \, dx$$

$$= 0 + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin^2 x \, dx \quad \checkmark$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$= (n-1) \left[\int_0^{\pi/2} \cos^{n-2} x \, dx - \int_0^{\pi/2} \cos^n x \, dx \right]$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{(n-1)}{n} I_{n-2} \quad \checkmark$$

$$I_3 = \frac{2}{3} I_1$$

$$= \frac{2}{3} \int_0^{\pi/2} \cos x \, dx$$

$$= \frac{2}{3} [\sin x]_0^{\pi/2}$$

$$= \frac{2}{3} [1 - 0]$$

$$= \frac{2}{3} \quad \checkmark$$

Q.2 (a)

$$\delta V = \pi (y+1)^2 \delta x$$

$$\delta V = \pi (\cos x + 1)^2 \delta x$$

$$V = \pi \int_0^{\pi} (\cos x + 1)^2 dx$$

$$= \pi \int_0^{\pi} (\cos^2 x + 2 \cos x + 1) dx$$

$$= \pi \int_0^{\pi} \left(\frac{1}{2} (1 + \cos 2x) + 2 \cos x + 1 \right) dx$$

$$= \pi \left[\frac{3x}{2} + \sin 2x + 2 \sin x \right]_0^{\pi}$$

$$= \pi \left\{ \left[\frac{3\pi}{2} + 0 + 0 \right] - [0] \right\}$$

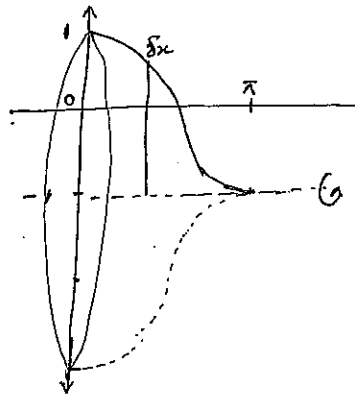
$$= \frac{3\pi^2}{2} \text{ units}^3$$

Now 1 cubic unit = $5 \times 5 \times 5 = 125 \text{ cm}^3$

$$\therefore V = \frac{3\pi^2}{2} \times 125$$

$$\doteq 589 \text{ cm}^3$$

$$\doteq 589 \text{ mL}$$



(b)

$$A = \pi (\text{outer radius})^2 - \pi (\text{inner radius})^2$$

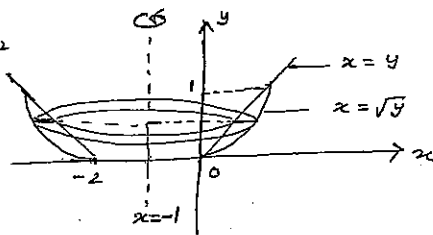
$$= \pi (1 + \sqrt{y})^2 - \pi (1 + y)^2$$

$$= \pi (2\sqrt{y} - y - y^2)$$

$$V = \pi \int_0^1 (2\sqrt{y} - y - y^2) dy$$

$$= \pi \left[\frac{4y^{3/2}}{3} - \frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

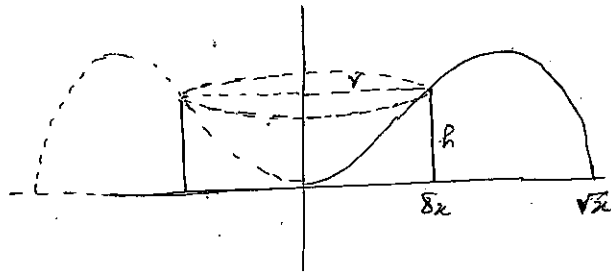
$$= \frac{\pi}{2}$$



(c)

$$\delta V = 2\pi r h \delta x$$

$$V = 2\pi \int_0^{\sqrt{\pi}} x \sin(x^2) dx$$



$$\begin{aligned} u &= x^2 \\ du &= 2x dx \\ \frac{du}{2x} &= dx \end{aligned} \quad \begin{aligned} \frac{dy}{dx} &= \sin(x^2) \\ y &= -\frac{\cos(x^2)}{2x} \end{aligned}$$

$$x=0 \quad u=0$$

$$x=\sqrt{\pi} \quad u=\pi$$

$$V = 2\pi \int_0^{\sqrt{\pi}} x \sin u \frac{du}{2x}$$

$$= \pi \int_0^{\pi} \sin u du$$

$$= \pi [-\cos u]_0^{\pi}$$

$$= \pi [-\cos \pi + \cos 0]$$

$$= \pi$$

(d)

Volume of cross-section:

$$\delta V = \frac{8y^2}{3} \delta x$$

$$\delta V = \frac{8}{3} (9 - x^2)^2 \delta x$$

$$V = \frac{8}{3} \int_{-3}^3 (9 - x^2)^2 dx \quad \checkmark$$

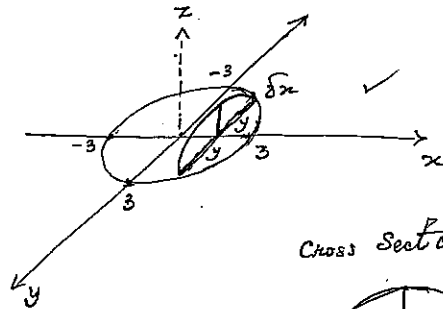
$$= \frac{8}{3} \left[9x - \frac{x^3}{3} \right]_{-3}^3$$

$$= \frac{8}{3} [(27 - 9) - (-27 + 9)]$$

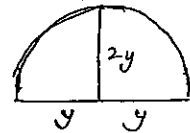
$$= \frac{8}{3} [18 + 18]$$

$$= 8 \times 12$$

$$= 96 \text{ cubic units.} \quad \checkmark$$



Cross Section



Simpson Rule.

$$A = \frac{y}{3} \{ 0 + 4(2y) + 0 \}$$

$$= \frac{8y^2}{3} \quad \checkmark$$